微积分 A1 第 3 次习题课答案　函数极限

1. 试求下列极限:

(1) $\lim\limits_{x\to 0}\dfrac{\cos 3x-\cos 7x}{x^2}$　　(2) $\lim\limits_{x\to 0}\dfrac{1-\sqrt{\cos x}}{x-x\cos\sqrt{x}}$

(3) $\lim\limits_{x\to 0}\dfrac{3\sin x+x^2\cos(1/x)}{(1+\cos x)\ln(1+x)}$　　(4) $\lim\limits_{x\to 0}\dfrac{\sin 2x+2\arctan 3x+3x^2}{\ln(1+3x+\sin^2 x)+xe^x}$

(5) $\lim\limits_{x\to+\infty}x\left[\left(1+\dfrac{a}{x}\right)^{1+1/x}-x^{-1/[x(x+a)]}\right]$　　(6) $\lim\limits_{x\to+\infty}\left(\cos\dfrac{1}{x}+\sin\dfrac{1}{x}\right)^x$

(7) $\lim\limits_{x\to 0}\left(\sqrt{1+x}-x\right)^{1/x}$　　(8) $\lim\limits_{x\to 0}\left[\ln(e+x)\right]^{\cot x}$

(9) $\lim\limits_{x\to 0}\left(\dfrac{xe^x+1}{x\pi^x+1}\right)^{1/x^2}$　　(10) $\lim\limits_{x\to 0}\left(\dfrac{a^{x+1}+b^{x+1}}{a+b}\right)^{1/x}\ (a,b>0)$

证明：（1）$\lim\limits_{x\to 0}\dfrac{\cos 3x-\cos 7x}{x^2}=\lim\limits_{x\to 0}\dfrac{1-\cos 7x}{x^2}-\lim\limits_{x\to 0}\dfrac{1-\cos 3x}{x^2}$

$$=\lim_{x\to 0}\frac{1-\cos 7x}{\frac{1}{2}(7x)^2}\cdot\lim_{x\to 0}\frac{\frac{1}{2}(7x)^2}{x^2}-\lim_{x\to 0}\frac{1-\cos 3x}{\frac{1}{2}(3x)^2}\cdot\lim_{x\to 0}\frac{\frac{1}{2}(3x)^2}{x^2}=\frac{49}{2}-\frac{9}{2}=20.$$

（2）$\lim\limits_{x\to 0}\dfrac{1-\sqrt{\cos x}}{x-x\cos\sqrt{x}}=\lim\limits_{x\to 0}\dfrac{1-e^{\frac{1}{2}\ln\cos x}}{x(1-\cos\sqrt{x})}$

$$=\frac{1}{2}\lim_{x\to 0}\frac{1-e^{\frac{1}{2}\ln\cos x}}{\frac{1}{2}\ln\cos x}\cdot\lim_{x\to 0}\frac{\ln(1+\cos x-1)}{\cos x-1}\cdot\lim_{x\to 0}\frac{\cos x-1}{\frac{1}{2}x^2}\cdot\lim_{x\to 0}\frac{\frac{1}{2}x}{(1-\cos\sqrt{x})}$$

$$=\frac{1}{2}\cdot(-1)\cdot 1\cdot(-1)\cdot 1=\frac{1}{2}.$$

（3）$\lim\limits_{x\to 0}\dfrac{3\sin x+x^2\cos(1/x)}{(1+\cos x)\ln(1+x)}=\lim\limits_{x\to 0}\dfrac{1}{(1+\cos x)}\cdot\lim\limits_{x\to 0}\dfrac{x}{\ln(1+x)}\cdot\lim\limits_{x\to 0}\dfrac{3\sin x+x^2\cos(1/x)}{x}$

$$=\frac{1}{2}\left(\lim_{x\to 0}\frac{3\sin x}{x}+\lim_{x\to 0}x\cos(1/x)\right)=\frac{1}{2}(3+0)=\frac{3}{2}.$$

（4） $\lim\limits_{x\to 0}\dfrac{\sin 2x+2\arctan 3x+3x^2}{\ln(1+3x+\sin^2 x)+xe^x}=\lim\limits_{x\to 0}\dfrac{\dfrac{\sin 2x}{x}+2\dfrac{\arctan 3x}{x}+3x}{\dfrac{\ln(1+3x+\sin^2 x)}{x}+e^x}$

$$=\frac{\lim\limits_{x\to 0}\dfrac{\sin 2x}{x}+2\lim\limits_{x\to 0}\dfrac{\arctan 3x}{x}+\lim\limits_{x\to 0}3x}{\lim\limits_{x\to 0}\dfrac{\ln(1+3x+\sin^2 x)}{x}+\lim\limits_{x\to 0}e^x}$$

$$=\frac{2+6}{1+\lim\limits_{x\to 0}\dfrac{\ln(1+3x+\sin^2 x)}{3x+\sin^2 x}\cdot\lim\limits_{x\to 0}\dfrac{3x+\sin^2 x}{x}}=\frac{8}{1+1\cdot 3}=2.$$

（5） $\lim\limits_{x\to+\infty}x\left[\left(1+\dfrac{a}{x}\right)^{1+1/x}-x^{-1/[x(x+a)]}\right]$

$$=\lim_{x\to+\infty}\left[x\left(e^{(1+\frac{1}{x})\ln\left(1+\frac{a}{x}\right)}-1\right)-x\left(e^{\frac{-\ln x}{x(x+a)}}-1\right)\right]$$

$$=\lim_{x\to+\infty}x\left(e^{(1+1/x)\ln(1+a/x)}-1\right)-\lim_{x\to+\infty}x\left(e^{\frac{-\ln x}{x(x+a)}}-1\right)$$

$$=\lim_{x\to+\infty}x(1+\frac{1}{x})\ln\left(1+\frac{a}{x}\right)-\lim_{x\to+\infty}\frac{-x\ln x}{x(x+a)}$$

$$=\lim_{x\to+\infty}x(1+\frac{1}{x})\frac{a}{x}-0$$

$$=a.$$

注：等价无穷小因子替换，简化计算过程。

（6）**解法一：** $\lim\limits_{x\to+\infty}\left(\cos\dfrac{1}{x}+\sin\dfrac{1}{x}\right)^x=\lim\limits_{x\to+\infty}\left(1+\cos\dfrac{1}{x}+\sin\dfrac{1}{x}-1\right)^{\frac{1}{\cos\frac{1}{x}+\sin\frac{1}{x}-1}\cdot x\left(\cos\frac{1}{x}+\sin\frac{1}{x}-1\right)}$

$$=e^{\lim\limits_{x\to+\infty}x\left(\cos\frac{1}{x}+\sin\frac{1}{x}-1\right)}=e^{\lim\limits_{x\to+\infty}x\left(\cos\frac{1}{x}-1\right)+\lim\limits_{x\to+\infty}\frac{\sin(1/x)}{1/x}}=e^{1+\lim\limits_{x\to+\infty}x\left(-\frac{1}{2x^2}\right)}=e.$$

解法二： $\lim\limits_{x\to+\infty}\left(\cos\dfrac{1}{x}+\sin\dfrac{1}{x}\right)^x=\lim\limits_{x\to+\infty}e^{x\ln\left(\cos\frac{1}{x}+\sin\frac{1}{x}\right)}=\exp\left\{\lim\limits_{x\to+\infty}x\ln\left(\cos\dfrac{1}{x}+\sin\dfrac{1}{x}\right)\right\}$

$$=\exp\left\{\lim_{x\to+\infty}x\left(\cos\frac{1}{x}+\sin\frac{1}{x}-1\right)\right\}=\exp\left\{\lim_{t\to 0^+}\frac{\cos t+\sin t-1}{t}\right\}$$

$$=\exp\left\{\lim_{t\to 0^+}\frac{\cos t-1}{t}+\lim_{t\to 0^+}\frac{\sin t}{t}\right\}=e.$$

（7）$\lim\limits_{x\to 0}\left(\sqrt{1+x}-x\right)^{1/x}=\lim\limits_{x\to 0}(1+x)^{\frac{1}{2x}}\left(1-\frac{x}{\sqrt{1+x}}\right)^{1/x}=e^{1/2}\lim\limits_{x\to 0}\left(1-\frac{x}{\sqrt{1+x}}\right)^{1/x}$

$$=e^{1/2}\cdot e^{\lim\limits_{x\to 0}\frac{1}{x}\ln\left(1-\frac{x}{\sqrt{1+x}}\right)}=e^{1/2}\cdot e^{\lim\limits_{x\to 0}\frac{-x}{x\sqrt{1+x}}}=e^{-1/2}.$$

（8）解法一：$\lim\limits_{x\to 0}\left(\ln(e+x)\right)^{\cot x}=\lim\limits_{x\to 0}\left(1+\ln(1+\frac{x}{e})\right)^{\cot x}$

$$=\lim_{x\to 0}\left(\left(1+\ln(1+\frac{x}{e})\right)^{\frac{1}{\ln(1+\frac{x}{e})}}\right)^{\cot x\cdot\ln(1+\frac{x}{e})}=\lim_{x\to 0}\left(\left(1+\ln(1+\frac{x}{e})\right)^{\frac{1}{\ln(1+\frac{x}{e})}}\right)^{\lim\limits_{x\to 0}\cot x\cdot\ln(1+\frac{x}{e})}$$

$$=\exp\left\{\lim_{x\to 0}\left(\cot x\cdot\ln(1+x/e)\right)\right\}=\exp\left\{\lim_{x\to 0}\left(\frac{x}{\tan x}\cdot\frac{\ln(1+x/e)}{x/e}\cdot\frac{1}{e}\right)\right\}=e^{1/e}.$$

解法二：$\lim\limits_{x\to 0}\left(\ln(e+x)\right)^{\cot x}=\lim\limits_{x\to 0}\left(1+\ln(1+\frac{x}{e})\right)^{\cot x}=\lim\limits_{x\to 0}\exp\left\{\cot x\ln\left(1+\ln(1+\frac{x}{e})\right)\right\}$

$$=\exp\left\{\lim_{x\to 0}\cot x\ln\left(1+\ln(1+\frac{x}{e})\right)\right\}=\exp\left\{\lim_{x\to 0}\frac{1}{x}\ln(1+\frac{x}{e})\right\}=\exp\left\{\lim_{x\to 0}(\frac{1}{x}\cdot\frac{x}{e})\right\}=e^{1/e}.$$

（9）$\lim\limits_{x\to 0}\left(\frac{xe^x+1}{x\pi^x+1}\right)^{1/x^2}=\lim\limits_{x\to 0}\left(1+\frac{x(e^x-\pi^x)}{x\pi^x+1}\right)^{1/x^2}=\lim\limits_{x\to 0}\exp\left\{\frac{1}{x^2}\ln\left(1+\frac{x(e^x-\pi^x)}{x\pi^x+1}\right)\right\}$

$$=\exp\left\{\lim_{x\to 0}\frac{1}{x^2}\ln\left(1+\frac{x(e^x-\pi^x)}{x\pi^x+1}\right)\right\}=\exp\left\{\lim_{x\to 0}\left(\frac{1}{x^2}\cdot\frac{x(e^x-\pi^x)}{x\pi^x+1}\right)\right\}$$

$$=\exp\left\{\lim_{x\to 0}\left(\frac{\pi^x}{x\pi^x+1}\cdot\frac{(e/\pi)^x-1}{x}\right)\right\}=e^{\ln(e/\pi)}=\frac{e}{\pi}.$$

（10）$\lim\limits_{x\to 0}\left(\frac{a^{x+1}+b^{x+1}}{a+b}\right)^{1/x}=\lim\limits_{x\to 0}\left(1+\frac{a(a^x-1)+b(b^x-1)}{a+b}\right)^{1/x}$

$$=\exp\left\{\lim_{x\to 0}\frac{1}{x}\ln\left(1+\frac{a(a^x-1)+b(b^x-1)}{a+b}\right)\right\}=\exp\left\{\lim_{x\to 0}\frac{a(a^x-1)+b(b^x-1)}{(a+b)x}\right\}$$

$$=\exp\left\{\frac{1}{a+b}\lim_{x\to 0}\left(\frac{a(a^x-1)}{x}+\frac{b(b^x-1)}{x}\right)\right\}=e^{\frac{a\ln a+b\ln b}{(a+b)}}=\left(a^ab^b\right)^{\frac{1}{a+b}}.$$

2. 已知 $\lim\limits_{x\to0^+}\dfrac{\ln(1+\dfrac{f(x)}{\tan x})}{2^x-1}=1$, 求 $\lim\limits_{x\to0^+}\dfrac{f(x)}{x^2}$.

解： $\lim\limits_{x\to0^+}\dfrac{\ln(1+\dfrac{f(x)}{\tan x})}{x}=\lim\limits_{x\to0^+}\dfrac{\ln(1+\dfrac{f(x)}{\tan x})}{2^x-1}\cdot\lim\limits_{x\to0^+}\dfrac{2^x-1}{x}=\ln 2,$

$\lim\limits_{x\to0^+}\ln(1+\dfrac{f(x)}{\tan x})=0,$ $\lim\limits_{x\to0^+}\dfrac{f(x)}{\tan x}=0,$

$$\lim_{x\to0^+}\frac{f(x)}{x^2}=\lim_{x\to0^+}\frac{\tan x}{x}\cdot\lim_{x\to0^+}\frac{\dfrac{f(x)}{\tan x}}{\ln(1+\dfrac{f(x)}{\tan x})}\cdot\lim_{x\to0^+}\frac{\ln(1+\dfrac{f(x)}{\tan x})}{x}=1\cdot1\cdot\ln 2=\ln 2.$$

3. 设 $0<m\le\dfrac{f(x)}{x^\alpha}\le M$, $\lim\limits_{x\to0^+}g(x)=0$, $\lim\limits_{x\to0^+}g(x)\ln x^\alpha=l$, 求证：$\lim\limits_{x\to0^+}f(x)^{g(x)}=e^l$.

证明： $0<m\le\dfrac{f(x)}{x^\alpha}\le M$, 则 $\ln\dfrac{f(x)}{x^\alpha}$ 有界. 又 $\lim\limits_{x\to0^+}g(x)=0$, 则

$$\lim_{x\to0^+}g(x)\ln\frac{f(x)}{x^\alpha}=0.$$

$$\lim_{x\to0^+}f(x)^{g(x)}=\lim_{x\to0^+}e^{g(x)\ln f(x)}=\lim_{x\to0^+}e^{g(x)\ln x^\alpha+g(x)\ln\frac{f(x)}{x^\alpha}}=e^l.\ \square$$

4. 当 $x\to+\infty$ 时, $f(x)=1-\cos(1-\cos\dfrac{1}{x})$ 与 αx^β 是等价无穷小. 求 α,β.

解： 当 $x\to+\infty$ 时,

$$f(x)=1-\cos(1-\cos\frac{1}{x})\sim\frac{1}{2}(1-\cos\frac{1}{x})^2\sim\frac{1}{2}\left(\frac{1}{2x^2}\right)^2=\frac{1}{8x^4},$$

因此, $\alpha=\dfrac{1}{8},\beta=-4.\ \square$

5. 设常数 $a_1,a_2,\cdots,a_n$ 满足 $a_1+a_2+\cdots+a_n=0$, 计算 $\lim\limits_{x\to\infty}\sum\limits_{k=1}^{n}a_k\sin\sqrt{x+k}$.

解： $a_1+a_2+\cdots+a_n=0$, 则 $\sum_{k=1}^{n}a_k\sin\sqrt{x}=0,\forall x\in\mathbb{R}$. 于是

$$\left|\sum_{k=1}^{n}a_k\sin\sqrt{x+k}\right|=\left|\sum_{k=1}^{n}a_k\sin\sqrt{x+k}-\sum_{k=1}^{n}a_k\sin\sqrt{x}\right|\le\sum_{k=1}^{n}|a_k|\left|\sin\sqrt{x+k}-\sin\sqrt{x}\right|$$

$$\le\sum_{k=1}^{n}|a_k|\left|\sqrt{x+k}-\sqrt{x}\right|=\sum_{k=1}^{n}\frac{|ka_k|}{\sqrt{x+k}+\sqrt{x}}\to 0(x\to\infty).$$

因此 $\lim_{x\to\infty}\sum_{k=1}^{n}a_k\sin\sqrt{x+k}=0.$

6. f 在$(0,+\infty)$上单调, $\lim_{x\to+\infty}\frac{f(2x)}{f(x)}=1,a>0$, 则 $\lim_{x\to+\infty}\frac{f(ax)}{f(x)}=1.$

证明： 对任意正整数 n, 有

$$\lim_{x\to+\infty}\frac{f(2^nx)}{f(x)}=\lim_{x\to+\infty}\frac{f(2^nx)}{f(2^{n-1}x)}\cdot\frac{f(2^{n-1}x)}{f(2^{n-2}x)}\cdots\cdots\frac{f(2x)}{f(x)}=1.$$

若$a\ge 1$, 则$\exists n>0,s.t.\ 2^0\le a<2^n$. f 单调, 不妨设f 单增, 则有

$$1=\frac{f(2^0x)}{f(x)}\le\frac{f(ax)}{f(x)}\le\frac{f(2^nx)}{f(x)},\quad\forall x>1.$$

由夹挤原理, $\lim_{x\to+\infty}\frac{f(ax)}{f(x)}=1.$

若$0<a<1$, 则 $\lim_{x\to+\infty}\frac{f(ax)}{f(x)}=\lim_{t\to+\infty}\frac{f(t)}{f(t/a)}=1.$□

7. $\lim_{x\to0}f(x)=0,\lim_{x\to0}\frac{f(x)-f(x/2)}{x}=0$, 则$\lim_{x\to0}\frac{f(x)}{x}=0.$

证明： $\forall\varepsilon>0$, 由$\lim_{x\to0}\frac{f(x)-f(x/2)}{x}=0,\exists\delta>0,s.t.$

$$|f(x)-f(x/2)|<\varepsilon|x|,\ \forall 0<|x|<\delta.$$

于是

$$|f(x)|\le\sum_{k=1}^{n}\left|f(\frac{x}{2^{k-1}})-f(\frac{x}{2^k})\right|+\left|f(\frac{x}{2^n})\right|$$

$$\le \sum_{k=1}^{n}\frac{\varepsilon|x|}{2^{k-1}}+\left|f(\frac{x}{2^n})\right|<2\varepsilon|x|+\left|f(\frac{x}{2^n})\right|,\quad \forall 0<|x|<\delta.$$

令$n\to+\infty$,由$\lim\limits_{x\to 0}f(x)=0$得

$$|f(x)|\le 2\varepsilon|x|,\quad \forall 0<|x|<\delta.$$

故$\lim\limits_{x\to 0}\dfrac{f(x)}{x}=0$.□

8. 若$\forall x\in(-1,1)$,有$\sum\limits_{k=1}^{n}|a_k\sin kx|\le|\sin x|$,求证:$\left|\sum\limits_{k=1}^{n}ka_k\right|\le 1$.

证明：$\lim\limits_{x\to 0}\dfrac{\sum\limits_{k=1}^{n}|a_k\sin kx|}{|\sin x|}=\lim\limits_{x\to 0}\sum\limits_{k=1}^{n}\left|\dfrac{a_k\sin kx}{\sin x}\right|=\sum\limits_{k=1}^{n}\left|\lim\limits_{x\to 0}\dfrac{a_k\sin kx}{\sin x}\right|=\sum\limits_{k=1}^{n}|ka_k|$.而由题意，有

$$\frac{\sum\limits_{k=1}^{n}|a_k\sin kx|}{|\sin x|}\le 1,\forall 0<|x|<1.$$

令$x\to 0$,得$\sum\limits_{k=1}^{n}|ka_k|\le 1$.□

有了函数极限后，计算数列极限有了新方法：可以借助函数极限来求数列极限：

9. 试求下列极限:

(1) $\lim\limits_{n\to+\infty}\left(\dfrac{2+\sqrt[n]{a}}{3}\right)^{2n-2}$ $(a>0)$　　(2) $\lim\limits_{n\to+\infty}n(\sqrt[n]{x}-\sqrt[2n]{x})$ $(x>0)$

(3) $\lim\limits_{n\to+\infty}\dfrac{1}{n}\sum\limits_{k=1}^{n}e^{k/n}$　　(4) $\lim\limits_{n\to+\infty}n^p\sin(\sqrt{2}+1)^n\pi$ $(p>0)$

解：（1） $\lim\limits_{n\to+\infty}\left(\dfrac{2+\sqrt[n]{a}}{3}\right)^{2n-2}=\lim\limits_{n\to+\infty}\left(1+\dfrac{\sqrt[n]{a}-1}{3}\right)^{\frac{3}{\sqrt[n]{a}-1}\cdot\frac{2(n-1)(\sqrt[n]{a}-1)}{3}}=\lim\limits_{n\to+\infty}e^{\frac{2(n-1)(\sqrt[n]{a}-1)}{3}}$

$$=\lim_{n\to+\infty}e^{\frac{2(n-1)\ln a}{3n}\cdot\frac{\sqrt[n]{a}-1}{\frac{1}{n}\ln a}}=e^{\frac{2}{3}\ln a}=a^{2/3}.$$

（2） $\lim\limits_{n\to+\infty}n(\sqrt[n]{x}-\sqrt[2n]{x})=\lim\limits_{n\to+\infty}\sqrt[2n]{x}\cdot\lim\limits_{n\to+\infty}n(\sqrt[2n]{x}-1)=\lim\limits_{n\to+\infty}\dfrac{\ln x}{2}\cdot\dfrac{\sqrt[2n]{x}-1}{\frac{1}{2n}\ln x}=\dfrac{\ln x}{2}$.

法二：$\lim\limits_{n\to+\infty} n(\sqrt[n]{x}-\sqrt[2n]{x})=\lim\limits_{n\to+\infty}\dfrac{\sqrt[n]{x}-1}{1/n}-\lim\limits_{n\to+\infty}\dfrac{1}{2}\dfrac{\sqrt[2n]{x}-1}{1/(2n)}=\ln x-\dfrac{1}{2}\ln x=\dfrac{1}{2}\ln x.$

（3）$\lim\limits_{n\to+\infty}\dfrac{1}{n}\sum\limits_{k=1}^{n}e^{k/n}=\lim\limits_{n\to+\infty}\dfrac{e^{1/n}(e-1)}{n(e^{1/n}-1)}=\dfrac{e-1}{\lim\limits_{n\to+\infty}\dfrac{(e^{1/n}-1)}{1/n}}=e-1.$

（4）$(\sqrt{2}+1)^n+(1-\sqrt{2})^n$为整数,所以

$$\sin\left((\sqrt{2}+1)^n\pi+(1-\sqrt{2})^n\pi\right)=0,\quad \left|\sin(\sqrt{2}+1)^n\pi\right|=\left|\sin(1-\sqrt{2})^n\pi\right|.$$

$$\lim_{n\to+\infty}\left|n^p\sin(\sqrt{2}+1)^n\pi\right|=\lim_{n\to+\infty}\frac{\sin(\sqrt{2}-1)^n\pi}{(\sqrt{2}-1)^n\pi}\cdot\lim_{n\to+\infty}\frac{n^p\pi}{(\sqrt{2}+1)^n}=1\cdot 0=0.$$

$$\lim_{n\to+\infty}n^p\sin(\sqrt{2}+1)^n\pi=0.$$

10. 若$2a^{1/n}-b^{1/n}-c^{1/n}=o(1/n)(n\to\infty)$,试问正数$a,b,c$满足什么关系？

解： $$0=\lim_{n\to\infty}\frac{2a^{1/n}-b^{1/n}-c^{1/n}}{1/n}=\lim_{n\to\infty}\frac{2(a^{1/n}-1)}{1/n}-\lim_{n\to\infty}\frac{b^{1/n}-1}{1/n}-\lim_{n\to\infty}\frac{c^{1/n}-1}{1/n}$$

$$=2\ln a-\ln b-\ln c=\ln\frac{a^2}{bc}.$$

故$a^2=bc.$ □

11. 设$-1<a_0<1,a_n=\sqrt{\dfrac{1+a_{n-1}}{2}},b_n=4^n(1-a_n)$. 求$\lim\limits_{n\to\infty}b_n$.

解： 记$a_0=\cos\theta(0<\theta<\pi)$,则$a_1=\sqrt{\dfrac{1+\cos\theta}{2}}=\cos\dfrac{\theta}{2},a_n=\cos\dfrac{\theta}{2^n},$

$$b_n=4^n(1-a_n)=4^n(1-\cos\frac{\theta}{2^n})=2^{2n+1}\sin^2\frac{\theta}{2^{n+1}}\to\frac{\theta^2}{2}(n\to\infty).$$

12. 设$b_1>a_1>0,a_{n+1}=(a_n+b_n)/2,b_{n+1}=\sqrt{a_{n+1}b_n}$. 求$\lim\limits_{n\to\infty}a_n,\lim\limits_{n\to\infty}b_n$.

解： 记$a_1=b_1\cos\theta(0<\theta<\pi/2)$,归纳可证

$$a_{n+1}=b_1\cos\frac{\theta}{2}\cos\frac{\theta}{2^2}\cdots\cos\frac{\theta}{2^{n-1}}\cos^2\frac{\theta}{2^n},$$
$$b_{n+1}=b_1\cos\frac{\theta}{2}\cos\frac{\theta}{2^2}\cdots\cos\frac{\theta}{2^n}.$$

因此，
$$\lim_{n\to\infty} b_n = b_1 \lim_{n\to\infty} \frac{\frac{1}{2^n}\sin\theta}{\sin\frac{\theta}{2^n}} = \frac{b_1\sin\theta}{\theta},$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n \cos\frac{\theta}{2^n} = \lim_{n\to\infty} b_n = \frac{b_1\sin\theta}{\theta}. \square$$

13. 求 $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sqrt[3]{1+\frac{k}{n^2}}-1\right)$.

解： $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{k}{3n^2}=\lim\limits_{n\to\infty}\frac{n+1}{6n}=\frac{1}{6}$. $\forall 0<\varepsilon<1, \exists N_1\in\mathbb{N}$, 当$n>N_1$时, 有

$$\frac{1}{6}(1-\varepsilon)<\sum_{k=1}^{n}\frac{k}{3n^2}<\frac{1}{6}(1+\varepsilon).$$

对此$\varepsilon>0$, 因 $\lim\limits_{x\to 0}\frac{(1+x)^{1/3}-1}{x/3}=1$, $\exists\delta>0$, 当$0<x<\delta$时, 有

$$1-\varepsilon<\frac{(1+x)^{1/3}-1}{x/3}<1+\varepsilon.$$

对此$\delta>0$, $\exists N>N_1$, 使得$1/N<\delta$. 因此，当$n>N$时, 有

$$\frac{1}{6}-\frac{\varepsilon}{3}<\frac{(1-\varepsilon)^2}{6}<(1-\varepsilon)\sum_{k=1}^{n}\frac{k}{3n^2}<\sum_{k=1}^{n}\left(\sqrt[3]{1+\frac{k}{n^2}}-1\right)<(1+\varepsilon)\sum_{k=1}^{n}\frac{k}{3n^2}<\frac{(1+\varepsilon)^2}{6}<\frac{1}{6}+\frac{\varepsilon}{2},$$

由极限定义，有 $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sqrt[3]{1+\frac{k}{n^2}}-1\right)=\frac{1}{6}. \square$

注：以下解法是错误的。

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(\sqrt[3]{1+\frac{k}{n^2}}-1\right)=\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{k}{3n^2}+o(\frac{k}{n^2})\right)=\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{k}{3n^2}+o(\frac{1}{n^2})\right)$$

$$=\lim_{n\to\infty}\left(o(\frac{1}{n})+\sum_{k=1}^{n}\frac{k}{3n^2}\right)=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{3n^2}=\frac{1}{6}.$$

这样修改也是错误的：

$$\lim_{n\to\infty}\sum_{k=1}^{n}\left(\sqrt[3]{1+\frac{k}{n^2}}-1\right)=\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{k}{3n^2}+o(\frac{k}{n^2})\right)=\lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{k}{3n^2}+o(\frac{1}{n})\right)$$

$$=\lim_{n\to\infty}\left(o(1)+\sum_{k=1}^{n}\frac{k}{3n^2}\right)=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k}{3n^2}=\frac{1}{6}.$$

事实上， $\alpha \geq 1, n \to +\infty$时, n个$o(\frac{1}{n^{\alpha}})$之和不一定是$o(\frac{1}{n^{\alpha-1}})$. 例如：

$$x_{kn} = \frac{1}{(n-k)^{\alpha+1} + \frac{1}{k^n}}, \forall k, n \in \mathbb{N},$$

则任意给定 k, $x_{kn} = o(\frac{1}{n^{\alpha}}), n \to +\infty$. 但

$$\sum_{k=1}^{n} x_{kn} \geq x_{nn} = n^n \neq o(1), n \to +\infty.$$

另一个颠覆直观的结论：无穷多个无穷小量的乘积不一定是无穷小量。例如：

$$f_1(x) \triangleq \frac{1}{2^k}, x \in [k, k+1), k = 0, 1, 2, \cdots$$

则

$$f_1(x) = 1, x \in [0,1), \quad \text{且} \quad \lim_{x \to +\infty} f_1(x) = 0.$$

定义

$$f_2(x) \triangleq \begin{cases} 1 & x \in [0,1), \\ \dfrac{1}{f_1(x)} & x \in [1,2), \\ f_1(x) & x \in [2, +\infty). \end{cases}$$

则

$$f_1(x) f_2(x) = 1, x \in [0,2), \quad \text{且} \quad \lim_{x \to +\infty} f_2(x) = 0.$$

归纳定义

$$f_n(x) \triangleq \begin{cases} 1 & x \in [0, n-1), \\ \dfrac{1}{f_1(x) f_2(x) \cdots f_{n-1}(x)} & x \in [n-1, n), \\ f_1(x) & x \in [n, +\infty). \end{cases}$$

则

$$f_1(x) f_2(x) \cdots f_n(x) = 1, x \in [0, n), \quad \text{且} \quad \lim_{x \to +\infty} f_n(x) = 0.$$

无穷乘积定义为前 n 项乘积的极限，即

$$f_1(x) f_2(x) \cdots f_n(x) \cdots \triangleq \lim_{n \to +\infty} \left(f_1(x) f_2(x) \cdots f_n(x) \right), \forall x.$$

$\forall x > 0, \exists n \in \mathbb{N}, s.t. x \in [0, n)$, 由以上构造，有

$$f_1(x) f_2(x) \cdots f_n(x) = 1, \quad \text{且} \quad f_k(x) = 1, \forall k > n.$$

于是，

$$\prod_{n=1}^{\infty} f_n(x) \equiv 1.$$